

Toxicity Report – Acidic Bonding Hair Treatment Ingredients

This report summarises safety and toxicity information for each listed ingredient in this hair treatment. It is not a diagnosis or personalised medical advice. For individual concerns (allergies, pregnancy, existing skin disease, etc.), consult a doctor or dermatologist.

1. Summary Table

Ingredient	Main Role	Known Toxicity / Concerns at Cosmetic Levels	Practical Risk (Normal Adult Scalp/Hair Use)
Aqua (Water)	Solvent	No relevant toxicity concerns.	Very low
Cetearyl Alcohol	Fatty alcohol emollient, thickener	Generally non-irritating; rare contact allergy in sensitive people.	Very low
Glycerin	Humectant (binds water)	Very low toxicity; may sting on very damaged skin.	Very low
Behentrimonium Chloride	Cationic conditioning agent, emulsifier	CIR expert panel: safe as used in cosmetics; higher test concentrations can irritate skin and eyes.	Low-moderate (local irritation if sensitive).
Stearyl Alcohol	Fatty alcohol emollient, thickener	Similar to cetearyl alcohol; low toxicity, rare allergy.	Very low
Citric Acid	Alpha-hydroxy acid (AHA), pH adjuster	At higher strengths and low pH, AHAs can irritate skin and increase sun sensitivity; this is why the label warns about possible sunburn.	Low-moderate (irritation, photosensitivity with overuse or on sensitive skin).
Cetyl Esters	Wax-like emollient, texture agent	Low dermal absorption and low	Very low

		toxicity; rarely irritating.	
Sodium Citrate	Buffer (controls pH)	Inert at cosmetic levels; very low toxicity.	Very low
Isopropyl Alcohol	Solvent, helps product spread and dry	High concentrations can be drying and irritating; systemic toxicity only with much larger exposures than from normal cosmetic use.	Low (mainly dryness/irritation if scalp is already compromised).
Parfum (Fragrance)	Fragrance blend	Common cause of contact allergy and irritation; exact composition undisclosed.	Moderate for fragrance-sensitive users
Phenoxyethanol	Preservative	EU SCCS and other regulators: safe up to 1% in cosmetics with an adequate safety margin; main issues are irritation or allergic eczema in a minority.	Low-moderate (preservative irritation/allergy in susceptible people).
Polyquaternium-10	Film-forming, conditioning polymer	CIR safety assessment: safe in current cosmetic uses; not a human sensitiser at tested levels.	Very low
Polysorbate 20	Non-ionic surfactant, solubiliser	CIR: safe as used when formulated to be non-irritating; some animal data show possible sensitisation at high challenge doses.	Low-moderate (possible mild irritation in very sensitive skin).

Hydroxypropyl Guar	Thickener, conditioning polymer	Large polymer with minimal skin penetration; low toxicity and low irritation potential.	Very low
Limonene	Fragrance component (citrus)	Recognised fragrance allergen; oxidised limonene is particularly sensitising.	Moderate for perfume-allergic users
Linalool	Fragrance component (floral)	Recognised fragrance allergen; can form sensitising oxidation products on air exposure.	Moderate for perfume-allergic users

2. Detailed Notes by Ingredient Group

A. Fatty Alcohols, Conditioners and Base Ingredients

Water, cetearyl alcohol, stearyl alcohol, glycerin, cetyl esters, sodium citrate, polyquaternium-10 and hydroxypropyl guar form the main base and conditioning system. The fatty alcohols (cetearyl and stearyl) and cetyl esters give slip and thickness; unlike short-chain “drying” alcohols, they are generally non-irritating and have very low systemic toxicity, with rare cases of allergy.

Glycerin is a humectant with a long history of safe use; it can sting slightly on very damaged skin but is otherwise well tolerated.

Polyquaternium-10 and hydroxypropyl guar are large cationic and polysaccharide polymers that form a conditioning film on hair; the Cosmetic Ingredient Review (CIR) concluded that polyquaternium-10 is safe in present cosmetic uses and is not a human sensitizer at up to 2% in testing. Behentrimonium chloride is another cationic surfactant used for conditioning; CIR reviews find it safe at the low concentrations used in hair and skin products, though concentrated solutions can be irritant, especially to eyes.

B. Acids, Buffers and Solvents (AHA-related warning)

Citric acid is an alpha-hydroxy acid (AHA) used here primarily to adjust pH and contribute to the “acidic bonding” effect on hair. AHAs are well known to increase skin sensitivity to the sun at higher strengths and low pH, which is why the label warns about possible sunburn and suggests limiting sun exposure while using the product and for a week afterwards. In a rinse-off hair product, contact time with scalp skin is relatively short, but people with very sensitive or broken scalp skin may still notice tingling or irritation.

Sodium citrate is the salt form of citric acid and helps buffer the pH. Isopropyl alcohol acts as a volatile solvent to help the formula spread and then rinse; in high concentrations it can be drying or irritating, but systemic toxicity would require exposures much greater than those from normal cosmetic use.

C. Preservative and Surfactant

Phenoxyethanol is the primary preservative. The EU Scientific Committee on Consumer Safety (SCCS) and other expert bodies consider it safe at up to 1% in cosmetics, with a comfortable margin of safety based on dermal exposure data. At those levels, the main concern is local irritation or allergic contact dermatitis in a susceptible minority rather than systemic toxicity.

Polysorbate 20 is a non-ionic surfactant and solubiliser. CIR reviews have repeatedly concluded that polysorbates are safe as used in cosmetics when formulated to be non-irritating. Some animal sensitisation assays showed sensitisation potential at high challenge doses, but this has not translated into frequent human allergy at normal use levels.

D. Fragrance and Fragrance Allergens

“Parfum” represents a mixture of fragrance components that is not fully disclosed. In addition, the label names limonene and linalool, which are specific fragrance allergens that must be listed when present above very low thresholds. Both substances are widely used citrus/floral scent molecules whose oxidised forms are recognised contact allergens. For most users they simply provide scent, but for individuals with perfume allergy or chronic eczema, this group of ingredients is the main source of potential problems (itching, redness, rash).

E. Overall Assessment

From a toxicology perspective, this hair treatment does not contain ingredients that are regarded as highly toxic or carcinogenic at cosmetic concentrations. Regulatory and expert panels have specifically reviewed behentrimonium chloride, polyquaternium-10, polysorbate 20 and phenoxyethanol and concluded that they are safe as used in cosmetics when formulated to minimise irritation.

The realistic risks for normal adult users are local: scalp or skin irritation, dryness, and fragrance-related allergy. Because the formula includes an AHA (citric acid), there is a theoretical increase in sun sensitivity, mainly relevant if the product is used frequently on areas where the scalp is exposed. If you experience persistent burning, redness, flaking or rash after using this product, discontinue use and seek medical advice, taking this ingredient list with you.